INCH-POUND

MIL-T-27/138B
8 June 2012
SUPERSEDING
MIL-T-27/138A
13 March 1991

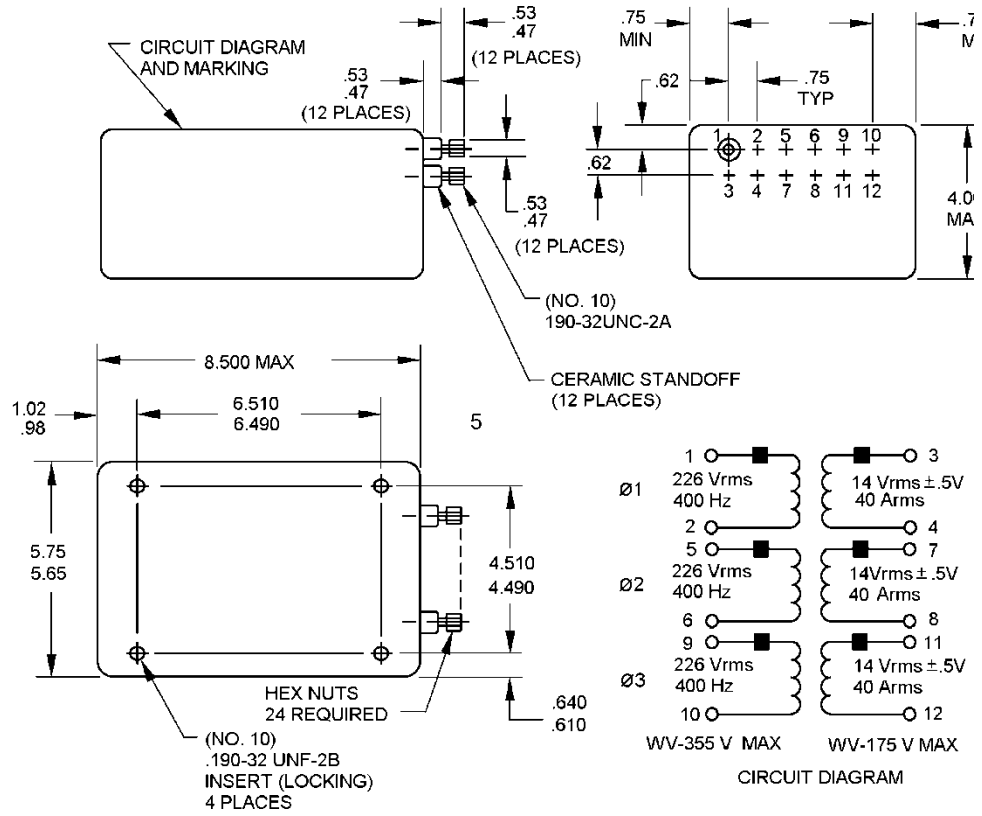
MILITARY SPECIFICATION SHEET

# TRANSFORMER, POWER, STEP-DOWN, 3-PHASE

Inactive for new design after
13 March 1991.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the product described herein shall consist of this document and the latest issue of specification MIL-PRF-27.




| Inches | mm |
|---|---|
| .190 | 4.83 |
| .47 | 11.9 |
| .53 | 13.5 |
| .610 | 15.49 |
| .62 | 15.7 |
| .640 | 16.3 |
| .75 | 19.0 |
| .93 | 24.9 |
| 1.02 | 25.9 |
| 4.490 | 114.05 |
| 4.510 | 114.44 |
| 5.65 | 143.5 |
| 5.75 | 146.0 |
| 6.490 | 164.8 |
| 6.510 | 165.34 |
| 8.500 | 215.90 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configuration.



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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary power: 585 watts each phase.

Primary voltage: (1-2) (5-6) (9-10): 226 volts rms ±5 percent, 400 Hz, 3-phase.

Secondary voltage: (3-4) (7-8) (11-12): 14 volts rms, 40 amperes rms ±.5 volt.

DC resistance:

(1-2) (5-6) (9-10): 0.5 ohm maximum.
(3-4) (7-8) (11-12): 0.1 ohm maximum.

Working voltage:

(1-2) (5-6) (9-10): 335 volts maximum.
(3-4) (7-8) (11-12): 175 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal.

Terminals: Ceramic standoff type.

Terminal height: 1.06 inch maximum.

Weight: 30 pounds maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds.

Dielectric withstanding voltage:

At sea level:

(1-2) (5-6) (9-10): 1,000 volts dc.
(3-4) (7-8) (11-12): 500 volts dc.

At barometric pressure:

(1-2) (5-6) (9-10): 900 volts dc.
(3-4) (7-8) (11-12): 455 volts dc.

Electrical characteristics:

No load: With 226 volts rms, 400 Hz, 3-phase across (1-2) (5-6) (9-10).

Voltage across (3-4) (7-8) (11-12): 17 volts maximum.

Rated load: with 226 volts rms, 400 Hz, 3 phase across (1-2) (5-6) (9-10), the voltage across (3-4) (7-8) (11-12) shall be 14 volts ±0.5.



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Polarity: Additive with terminals 2 and 3; 6 and 7; 10 and 11 connected.

Temperature rise: 45°C maximum with 226 volts rms ±5 percent, 400 Hz, 3 phase across (1-2) (5-6) (9-10) at an ambient temperature of 85°C.

Salt spray: The transformers shall be subjected to a 20 percent salt spray solution for 50 hours.

VERIFICATION:

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/138-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - ER
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-087)

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil .



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